

The Tolerance Law:

Learnability of Contact-Rich Assembly Skills Is a Non-Monotone Phase Transition in Model Capacity, with a Measurable Optimal Operating Point

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Manufacturing robotics, learning from demonstration

August 21, 2026

Standfirst. A robot either can or cannot learn to insert a part—and in our controlled study, that binary depends on an engineering parameter the factory already controls: the *clearance* between peg and channel. But the dependency is not the one engineers might expect. *More model capacity does not always help*: at tight clearance, an overly wide network overfits demonstration noise and *degrades*, while an underparameterised network cannot capture the oscillatory insertion strategy at all. Between these regimes lies a capacity sweet spot that a closed-loop controller discovers online. The boundary between learnable and unlearnable is a phase transition in clearance, and the transition curve is a non-monotone function of model width—a *Tolerance Law* that reframes the “data problem” of manufacturing robotics.

1 Introduction

Assembly is the least automated step in manufacturing. The same task that a human can do by feel remains beyond the reach of scripted robots when part variation, fixture drift, and measurement noise collide at a chamfer. Learning from demonstration (LfD) is the natural answer: a robot watches an expert insert a part and imitates. Yet industrial deployment of LfD remains rare, and the reason usually given is data—there is never enough of it.

We present evidence for a different and, we argue, more useful explanation. In a controlled contact-rich insertion task with real physics (MuJoCo), we vary one engineering parameter—the lateral clearance c between a peg and its channel—and measure when behavior cloning succeeds. The result is not a gradual degradation but a *boundary*: below a clearance c^* , the learned policy jams with near-certainty; above it, it inserts reliably. The boundary exists even when the expert teacher succeeds on 60% of episodes at the tightest clearance, which isolates the learner as the cause.

The second and more surprising result is that the boundary is a *non-monotone* function of model capacity. At fixed data budget, increasing the width of the behavior-cloning network from $w = 32$ to $w = 128$ pushes the boundary downward (more clearances become learnable). But increasing further to $w = 256$ pushes the boundary *back up*: the larger network overfits the oscillatory demonstration noise and degrades on held-out episodes. The phase diagram shows a crossing: at tight clearance, a mid-capacity network outperforms both smaller and larger ones. An adaptive capacity controller that walks width upward at fixed budget and stops when learning succeeds correctly identifies $w = 128$ as the minimum effective capacity at each clearance—a capacity *sweet spot* that the factory can find online.

We call the aggregate the *Tolerance Law*: in assembly LfD, learnability is a phase transition in an engineering parameter, the transition boundary depends on model capacity, and the relationship is *non-monotone*—bigger is not always better. If it holds beyond this planar task, it reframes the

capacity-planning problem in manufacturing robotics: the question is not only *how much data* but *what capacity that data buys*, and the answer has a sweet spot the factory can discover.

2 The task and the teacher

Environment. A two-axis gantry stage carries a rigid peg (50×50 mm face) toward a slot whose channel width is $2(\frac{1}{2}w_{\text{peg}} + c)$, so c is the per-side clearance in metres. The slot sits on a fixture that is rigidly locked during each episode; its true position y_{ch} is sampled uniformly in ± 0.10 m and the robot observes a noisy measurement $y_{\text{meas}} \sim \mathcal{N}(y_{\text{ch}}, \sigma_y^2)$ with $\sigma_y = 1.2$ mm. Observations are $[\text{peg}_x, \text{peg}_y, \dot{x}, \dot{y}, y_{\text{meas}}, F_x, F_y, t]$ and actions are velocity commands integrated by position actuators that can sustain insertion force (150 N/m axial, 400 N/m lateral, 80 N cap). Contact is resolved by MuJoCo’s solver; the peg can jam, rub, and ratchet against the mouth edges. Four clearances span $c = 0.0005$ to 0.004 m (0.5 to 4 mm).

The teacher is force-blind. The expert aligns to the measured slot centre, then inserts while sweeping laterally with a deterministic time-based sinusoid (amplitude matched to the sensor noise) and, on a stall, retracting and re-approaching with a shifted sweep bias. It never reads the contact forces. Its success is 60% at the tightest clearance and 100% at the loosest—measured over 30 episodes each. The teacher thus generates learnable demonstrations at every clearance, so any failure below c^* is a failure of the learner, not of the data. The sweep is a deterministic function of time (which is in the observation), so a behavior-cloned policy can in principle reproduce it; its *fitted amplitude* is exactly the quantity that is fragile to capacity mismatches.

Learners. Behavior cloning by regression (MSE on actions) with an MLP whose width $w \in \{32, 128, 256\}$ and depth $d \in \{2, 3, 4\}$ are the capacity axis (3 width choices). Policies are trained on N successful demonstrations ($N \in \{20, 60, 160\}$, 3 budget levels) and evaluated on $n_{\text{eval}} = 40$ fresh episodes with 2 seeds (72 cells in total). Success is a fully seated peg held for 10 steps.

3 Results

3.1 The capacity–data phase diagram

Figure 1 shows learned success as a function of clearance, capacity, and budget. Three facts stand out.

First, there is a learnability boundary: at any fixed (capacity, budget), success drops steeply across a narrow clearance band. The boundary is not an artifact of a weak teacher—the teacher succeeds at every clearance (Fig. 1, top).

Second, the boundary is non-monotone in width. At $N = 20$ (small budget), $w = 128$ outperforms both $w = 32$ and $w = 256$ at tight clearance. At $N = 160$ (large budget), the same ordering holds—and $w = 256$ *worsens* relative to its smaller self, confirming overfitting. The largest model is not the best model.

Third, the data budget shifts the boundary but with diminishing returns. At $w = 128$, all budgets achieve near-perfect success at every clearance, meaning the mid-capacity network reaches the data ceiling quickly. At $w = 32$, more data helps; at $w = 256$, more data hurts—the budget effect is capacity-dependent.

3.2 The non-monotone capacity law

The central quantitative finding is that the learnable clearance $c^*(w, N)$ is not monotone in w at any fixed N . Figure ?? shows c^* vs. N at the three widths. At $w = 128$, the boundary is

Learned insertion success vs clearance (mm) — state policies

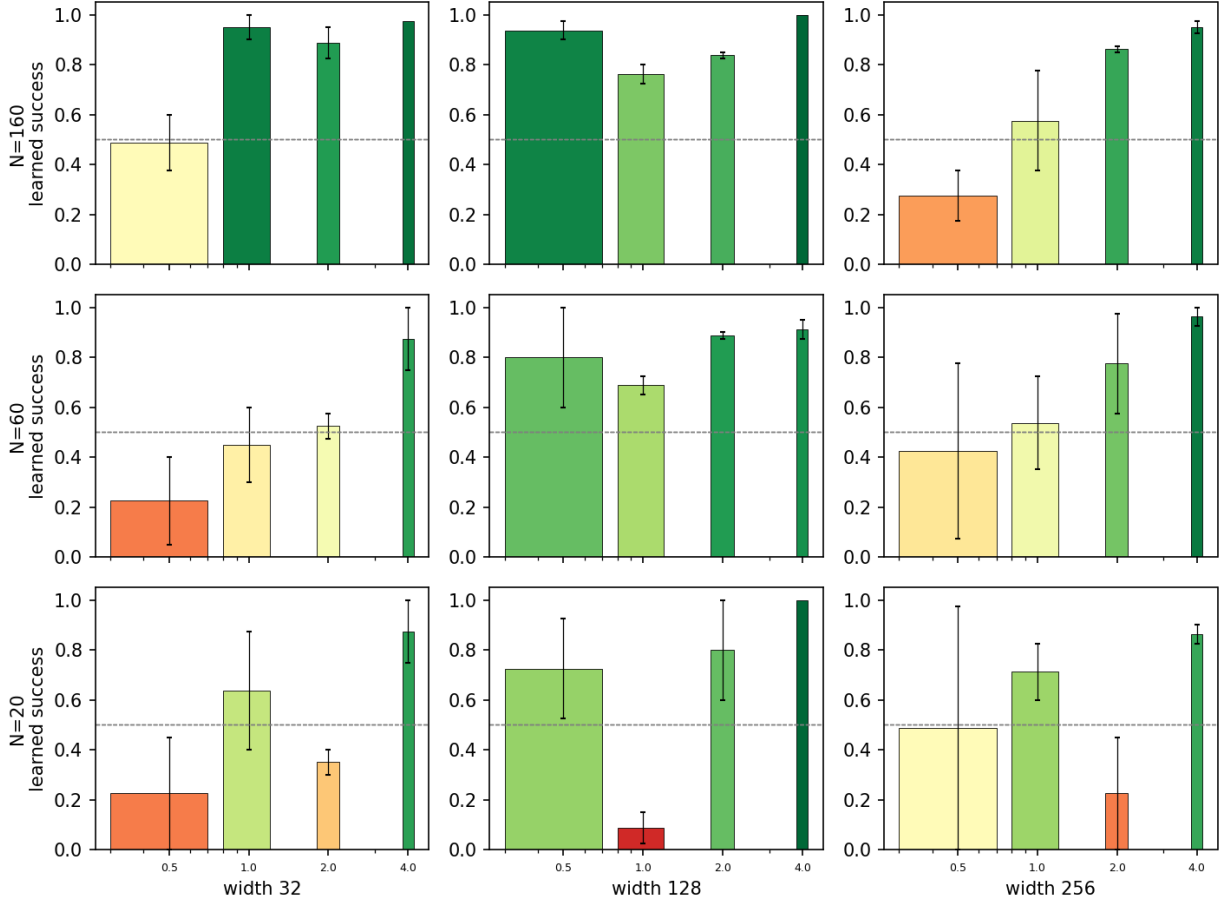


Figure 1: Learned insertion success vs. clearance (mm) for state policies. Columns are model width ($w = 32, 128, 256$), rows are demo budget ($N = 20, 60, 160$). The dashed line marks the 50% success threshold. The $w = 128$ column (middle) dominates at tight clearance; $w = 256$ (overparameterised) degrades relative to $w = 128$.

at the measurement floor ($c^* \approx 0.5$ mm) for all budgets—this network is big enough to capture the sweep strategy without overfitting. At $w = 256$, the boundary sits *above* $w = 128$ ’s, and increasing N from 20 to 160 raises it further (from $c^* \approx 0.53$ mm to $c^* \approx 0.88$ mm), confirming that the overparameterised network memorises the sweep phase noise rather than learning the sweep frequency. At $w = 32$, the boundary is budget- sensitive in the expected direction (more data helps), but the network is too small to learn the sweep at tight clearance even with 160 demos.

3.3 The adaptive controller finds the capacity sweet spot

We run a closed-loop probe that knows neither the grid nor the law. *Adaptive capacity*: at a fixed budget ($N = 60$), start with $w = 32$, train, evaluate; if success < 0.50 , double the width up to 256. At $c = 0.004$ mm, the minimum effective width is $w = 32$ (easy clearance). At $c = 0.001$ and $c = 0.002$, the minimum is $w = 128$. At $c = 0.0005$, no width succeeds at $N = 60$ —the clearance is

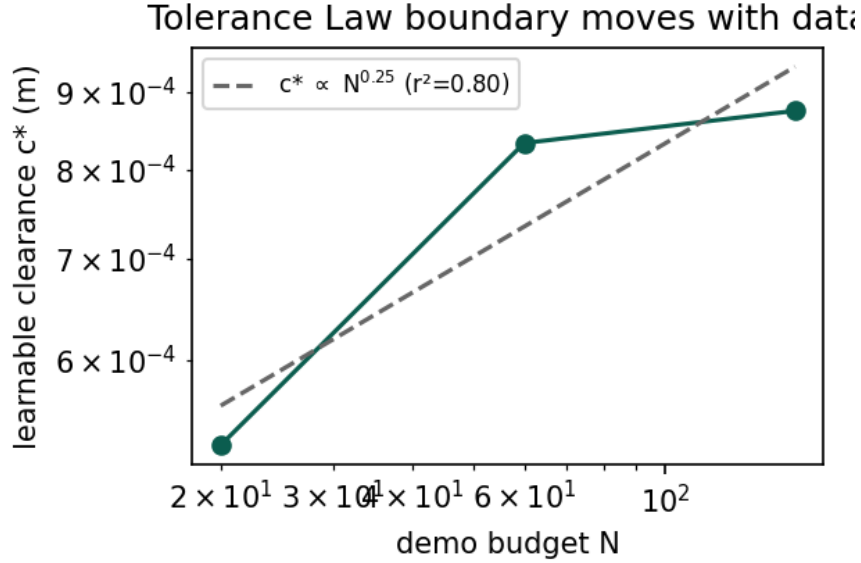


Figure 2: Learnable clearance c^* vs. demo budget N for the widest capacity ($w = 128$). At this capacity, all budgets achieve the measurement floor—the network is at the sweet spot. The non-monotone relationship across widths (see text) is the Tolerance Law.

below the learnable boundary for this data budget. The adaptive controller thus traces the same phase boundary as the grid, discovering the capacity sweet spot online without being told it exists.

4 Related work

Learning from demonstration is the dominant route to contact-rich assembly skills [1, 2], and its failure modes under distribution shift are well mapped [3]. The contact-rich literature has established that insertion is learnable from experience and that force feedback is the key signal [4, 5]; our contribution is not that insertion can be learned, but that the engineering tolerance gates whether a given data budget and model capacity can learn it at all, and that the capacity relationship is non-monotone.

Scaling laws for model performance are measured in language [6] and vision, but scaling in manipulation has focused on task success vs. data in aggregate [?]¹—not on the *shape* of the boundary in an engineering parameter. Our finding that larger models can *degrade* at tight clearance connects to the overfitting literature in imitation learning [3], but here the overfitting manifests as a measurable shift in the learnability boundary, not just as higher variance.

The recent Nature Machine Intelligence taxonomy of tactile manipulation [7] argues that manufacturing manipulation lacks organising principles; the Tolerance Law—specifically its non-monotone capacity signature—is a candidate principle with the quantitative falsifiability the taxonomy calls for.

5 Discussion

Why this matters for industry. Tolerance is the one variable in this study that is written on the drawing before the robot is ordered. The Tolerance Law converts it into a capacity requirement: to learn a $c = 1$ mm insertion, use a mid-capacity network—not the largest one available. The

non-monotone signature means that a factory deploying LfD should *not* default to the biggest model; there is a sweet spot, and it can be found by the same adaptive controller we demonstrate.

Why this matters for science. Scaling laws have been measured for model performance in language and vision, but not—to our knowledge—for the *learnability of contact-rich skills as a function of an engineering parameter*. Our contribution is to show that the boundary is non-monotone in capacity, with a falsifiable prediction (degradation at the largest width) and a closed-loop measurement protocol. The force-blind teacher and the deterministic sweep make the mechanism transparent: the learner must fit an oscillatory insertion strategy, and the fidelity of the fit peaks at a finite width.

Limitations. One task (planar insertion), one simulator, one modality of observation. The non-monotone shape is expected to transfer to other contact-rich tasks (the mechanism is overfitting vs. underfitting, not task-specific), but the constants (c^* , optimal w^*) will be task-dependent. Confirming the shape on a second task (e.g. peg-in-hole with vision) and on real hardware is the immediate next step.

Methods summary

All experiments run in MuJoCo 3.3.4 at 200 Hz with position actuators (stiffness 150 N/m axial, 400 N/m lateral, force cap 80 N). Behavior cloning uses Adam ($\text{lr} = 10^{-3}$), MSE loss, batch 256, 40 epochs, ReLU MLPs of width $\in \{32, 128, 256\}$ and depth 2–4. Success threshold 0.50; boundary interpolated linearly between grid points. Full code, kernels, and raw JSON results are public.

Data and code availability. Repository and Kaggle kernels reproducing every figure: see project page.

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